

VISHAL MUKUNDAN

+91 9048996477 | vishal.98m@gmail.com | Kannur, Kerala, India
[LinkedIn](#) | [GitHub](#)

RESEARCH INTERESTS

Semi-supervised and self-supervised representation learning, fewshot learning, domain adaptation and distribution shift in vision models with applications towards medical imaging and remote sensing.

EDUCATION

National Institute of Technology Karnataka (NITK)

Surathkal, Karnataka, India

Master of Technology in Signal Processing and Machine Learning | CGPA: **8.76 / 10**

Aug 2023 – Jul 2025

- *Relevant Coursework:* Deep Learning, Pattern Recognition & Machine Learning, Linear Algebra & Stochastic Processes, Optimization, High Performance Computing Architectures, Algorithms and Complexity

Government College of Engineering Kannur

Kerala, India

B.Tech (Hons) in Electrical and Electronics Engineering | CGPA: **8.56 / 10**

Jul 2016 – Jun 2020

PUBLICATIONS

[J1] Kevala, V.D.*, Mukundan, V.*, et al. "SARCDNet — an enhanced deep learning network for change detection from bi-temporal SAR images." *Scientific Reports* **16**, 1826 (2026). (*Equal Contribution)

[doi:10.1038/s41598-025-31488-y](https://doi.org/10.1038/s41598-025-31488-y)

RESEARCH EXPERIENCE

Computer Vision and Deep Learning Lab, NITK

May 2024 – Jun 2025

Research Assistant (M.Tech Thesis)

Surathkal, Karnataka, India

- Designed **SARCDNet**, a self-training deep learning architecture for binary change detection in multi-temporal SAR images, employing hierarchical FCM clustering to generate reliable pseudo-labels and eliminate the need for manual annotations.
- Achieved **4.26% improvement in F1 score** and **4.92% gain in Kappa coefficient** for flood detection on the Chao Lake Dataset over fully supervised baselines.
- Developed **SARSegNet**, a CNN-based land cover segmentation network leveraging multi-dilation feature extraction and residual attention mechanisms; improved IoU by 2.54% and F1 by 2.17% while using $25\times$ fewer parameters than reference models. Performed a TTA for uncertainty Analysis.

Government College of Engineering Kannur

Aug 2019 – Jun 2020

B.Tech Final Year Project — Wild Elephant Detection and Repelling System

Kannur, Kerala, India

- Developed a functional real-time object detection prototype using CCTV live feeds integrated with a Raspberry Pi edge device.
- Collected and annotated a custom infrared image(IR) dataset and trained a YOLOv3 detection model achieving **93% detection accuracy** on the test set.
- Implemented a low-latency inference pipeline to trigger visual and auditory deterrence mechanisms upon detection.
- Awarded funding from the **Rural Technology Development Fund, Kerala, India**, validating real-world societal impact of the research.

PROFESSIONAL EXPERIENCE

Auradine

Jul 2025 – Feb 2026

Software Intern

Bengaluru, Karnataka, India

- Engineered Python-based ETL data pipelines to aggregate and preprocess multi-source enterprise data into Snowflake, creating analytics and ML-ready datasets.
- Collaborated with cross-functional teams to design end-to-end data workflows supporting downstream machine learning applications.

TEACHING EXPERIENCE

National Institute of Technology Karnataka

Teaching Assistant, EC204: Digital Systems Design Laboratory

Jul 2024 – Dec 2024
Mangalore, Karnataka, India

- Supervised undergraduate lab sessions for a batch of 30 students, providing hands-on guidance in digital systems design experiments.
- Administered oral examinations to evaluate students' conceptual and practical understanding of digital systems.
- Graded lab records, assignments and practical assessments for the course.

KEY PROJECTS

Medical VQA Fine-Tuning

[HuggingFace](#)

- Fine-tuned **Qwen2.5-VL-3B-Instruct** on the VQA-RAD radiology benchmark using **QLoRA**; achieved 53.88% overall EM and 58.43% Token F1, a **+11.97% relative gain** over the base model.
- Engineered inference prompt to suppress verbose generation (enforcing 1–5 word outputs), directly improving exact-match accuracy.

3D Medical Image Segmentation Pipeline (BraTS 2023)

[GitHub](#)

- Built a 3D segmentation pipeline (**PyTorch**, **MONAI**, 3D U-Net) for BraTS 2023 glioma MRI, achieving **0.8889 mean Dice** across tumor sub-regions (WT: 0.9068, TC: 0.8964, ET: 0.8634) using per-channel weighted **DiceCELoss** to upweight the rare Enhancing Tumor class.
- Implemented **Monte Carlo Dropout** and 8-way **test-time augmentation** for uncertainty estimation, generating voxelwise variance and entropy maps alongside automated 4-panel visualizations for qualitative evaluation.

Transductive Few-Shot Learning for Abdominal CT under Domain Shift

[GitHub](#)

- Built a few-shot benchmark that compares an inductive baseline against three transductive methods on real abdominal CT slices drawn from axial, coronal, and sagittal planes.
- Ran a 200-episode 5-way benchmark showing the MDL-based transductive method improves 1-shot accuracy by 2 to 3 points over the inductive baseline on every plane, including a jump from 38.23% to 40.58% on out-of-domain coronal slices.
- Ran a robustness sweep and a cross-plane evaluation that surfaces a consistent domain shift effect, with baseline accuracy dropping from 66.30% on in-domain axial slices to 42.11% on out-of-domain sagittal slices.

TECHNICAL SKILLS

Machine Learning & AI: Computer Vision, NLP, Semi-Supervised Learning, Reinforcement Learning, LLM Fine-Tuning (LoRA, QLoRA), Retrieval-Augmented Generation (RAG, GraphRAG), Causal ML, SAR Image Processing

Programming Languages: SQL, Python, C++

Frameworks & Libraries: PyTorch, TensorFlow, Keras, HuggingFace Transformers, PEFT (Unsloth, TRL), LangChain, LangGraph, OpenCV, Scikit-Learn, Rasterio, Pandas, NumPy, SciPy, Matplotlib

Tools & Platforms: AWS (S3, EC2), Snowflake, PineconeDB, Git, Jupyter, VS Code, Evidently AI, Kaggle, ESA SNAP

HONOURS & AWARDS

- **Research Grant:** Investigator for a prototype funded by the *Rural Technology Development Fund, Kerala* for a "Wild Elephant Detection and Repelling System" (B.Tech Capstone Project, 2020).
- **2nd Prize:** State-Level Research Paper Presentation Competition, organized by the *Indian Society for Technical Education (ISTE) Student Convention*.